

30 Ans: C

Given half-life = 60 days & tin-124 is stable.

After one half-life: $N_{\text{Ant}} : N_{\text{Tin}} = 1:1$

After 2 half-lives:

$N_{\text{Ant}} : N_{\text{Tin}} = 1:3$ since $N_{\text{Ant}} = \frac{1}{4}$ Initial number of Antimony &
 $N_{\text{Tin}} = \frac{3}{4}$ of Initial number of Antimony

After 3 half-lives: $N_{\text{Ant}} : N_{\text{Tin}} = 1:7$ since $N_{\text{Ant}} = \frac{1}{8}$ Initial number of Antimony

$N_{\text{Tin}} = \frac{7}{8}$ of Initial number of Antimony

Thus $\frac{\text{number of tin-124 nuclei}}{\text{number of antimony-124 nuclei}} = 6$ occurs between 2 & 3 half-lives,
ie between 120 & 180 days. (C)

Alternatively, we can calculate the actual number of days as shown:

Since Tin is **stable**, $N_{\text{Ant}} + N_{\text{Tin}} = \text{initial number of Antimony, } N_0$

After time t : $N_{\text{tin}} = 6 N_{\text{Ant}}$, we can deduce $\frac{N_{\text{Ant}}}{N_0} = \frac{1}{7}$

Substitute into eqn $N = N_0 e^{-\lambda t}$ for Antimony, $t = 168$ days.